Sample Papers

MTH501

Final Term Examination – Spring 2006

Time Allowed: 150 Minutes

Question No. 1 **Marks : 2**

A linear transformation T : $R^2 \to R^2$ defined by $T(x) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_2 \\ x_1 \end{bmatrix}$, $u = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$, $v = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ then the image of u+v is

- $\begin{bmatrix} 4 \\ 6 \end{bmatrix}$
- $\begin{bmatrix} 4 \\ -6 \end{bmatrix}$
- $\begin{bmatrix} -4 \\ 6 \end{bmatrix}$
- $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$

Question No. 2 **Marks : 2**

In the set $S = \{ \begin{bmatrix} x - 2t \\ x + t \\ 3t \end{bmatrix}$; x , t in R $\}$ then its dimension is

- **1**
- **2**
- **0**
- **3**

Question No. 3 **Marks : 10**

Let $a_1 = \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix}$, $a_2 = \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}$, $b = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$ Determine whether b can be written as a linear combination of a_1 and a_2 ?

Note : In order to get full marks do all necessary steps

Question No. 4 **Marks : 10**

Determine whether the set $S = \{ v_1 , v_2 , v_3 \}$ form basis for R^3 ? Where $v_1 = \begin{bmatrix} 3 \\ 0 \\ -6 \end{bmatrix}$, $v_2 = \begin{bmatrix} -4 \\ 1 \\ 7 \end{bmatrix}$, $v_3 = \begin{bmatrix} -2 \\ 1 \\ 5 \end{bmatrix}$

Note : In order to get full marks do all necessary steps

Question No. 5 **Marks : 10**

Find the rank and nullity of the following matrix and show that the values obtained satisfy the dimension theorem

$$\begin{bmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & -4 & -5 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Note : In order to get full marks do all necessary steps

Question No. 6 **Marks : 2**

Let $A = \begin{bmatrix} 1 & -3 & -2 \\ -5 & 9 & 1 \end{bmatrix}$ and $u = \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix}$, then u is in

- Column space of A
- Null space of A
- Row space of A
- None of them

Question No. 7 **Marks : 2**

Let B = (b_1 , b_2) and C=(c_1 , c_2) for a vector space V such that $b_1 = 4c_1 + c_2$ and $b_2 = -6c_1 + c_2$. Suppose $x = 3b_1 + b_2$ and $[x]_B = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ then $[x]_C$ will be

- $\begin{bmatrix} 6 \\ 4 \end{bmatrix}$
- $\begin{bmatrix} -4 \\ 6 \end{bmatrix}$
- $\begin{bmatrix} -6 \\ 4 \end{bmatrix}$
- $\begin{bmatrix} -6 \\ -4 \end{bmatrix}$

Question No. 8 **Marks : 2**

Consider a basis B = (b_1 , b_2) for R^2 where $b_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $b_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and suppose x is in R^2 has the Coordinate vector $[x]_B = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$ then the value x is

- $\begin{bmatrix} -1 \\ 6 \end{bmatrix}$
- $\begin{bmatrix} -6 \\ 1 \end{bmatrix}$
- $\begin{bmatrix} -1 \\ -6 \end{bmatrix}$
- $\begin{bmatrix} 1 \\ 6 \end{bmatrix}$

Question No. 9 **Marks : 10**

Find the dimension of null space and column space of the matrix $\begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 5 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$

Note : In order to get full marks do all necessary steps

Question No. 10 **Marks : 10**

Find the characteristics polynomial , eigen value and eigenvector of the matrix

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \\ 4 & -4 & 5 \end{bmatrix}$$

Note : In order to get full marks do all necessary steps

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